

GROUP PROJECT

Build an Image Recognition System

From image classification to a business application

Team size: maximum 5 students

Minimum classes: 2 image categories

Objective

In this project, your team will design and develop a simple image classification system using a real-world image dataset. The project has two main goals:

- Build a working Machine Learning / Deep Learning model able to recognize different categories of images.
- Imagine how this technology could become part of a real business application.

You should therefore think both as data scientists and as a team developing a possible product or service.

Part 1 — Choose the Problem

Choose an image classification dataset from Roboflow Universe:

<https://universe.roboflow.com/classification>

Your dataset must contain at least two different classes.

Examples of possible classification problems

- Healthy vs diseased plants
- Different types of fruits
- Ripe vs unripe products
- Different types of clothing
- Different types of animals
- Different types of objects
- Product quality classification
- Different environments or locations
- Defective vs non-defective products

Describe your problem

- What does the dataset contain?
- What are the classes?
- How many images are available?
- What should the model learn to recognize?
- Why did your team choose this dataset?

Part 2 — Explore the Data

Before training the model, explore the dataset and show some example images from the different classes.

- Are the classes balanced?

- Are the images similar or very different from each other?
- Is the classification problem easy or difficult for a human?
- Are there images that could be ambiguous?
- What characteristics could help the model distinguish between the classes?

Think about what information is contained in the images and what patterns a neural network might learn.

Part 3 — Build the Image Classification Model

Create a model able to classify the images. You may use TensorFlow/Keras, PyTorch, Scikit-learn when appropriate, and Google Colab. You may either build a simple neural network architecture or use transfer learning with a pretrained model.

Data preparation

- Load the images
- Resize them if necessary
- Prepare the labels
- Normalize the data
- Divide the data into training, validation and/or test sets

Model

- What model did you use?
- What are its main layers or components?
- What is the input?
- What is the output?
- Which activation function is used for the final prediction?

Important: You do not need to build a very complex neural network. A simple model that you understand is preferable to a sophisticated model that you cannot explain.

Part 4 — Train and Evaluate the Model

Train your model and observe its performance. At minimum, report:

- Training accuracy
- Validation or test accuracy

You may also use additional metrics or visualizations such as:

- Loss
- Confusion matrix
- Precision
- Recall
- F1-score

Show some examples of correct and incorrect predictions. For at least a few images, show:

Image → True Class → Predicted Class

Questions to consider

- How well does the model work?
- Which classes are easier to recognize?
- Which classes are more frequently confused?
- Why do you think the model makes these mistakes?

- Is there evidence of overfitting?
- Would you trust the model on completely new images?

Part 5 — Test the System

Find or take some images that were not part of the original training dataset and use your model to classify them.

- An image that should be relatively easy to classify
- An image that could be difficult or ambiguous

Discuss whether the prediction makes sense and what this tells you about the model.

Part 6 — From a Model to a Business Application

Now imagine that your classifier is not simply a university exercise. Your team wants to turn it into part of a real product or service. Propose a business application based on your image recognition system. For example:

- Quality-control system in a factory
- Agricultural monitoring application
- Food selection system
- Retail application
- Recycling system
- Mobile application
- Visual inspection system
- Automated sorting system
- Decision-support tool

Do not simply describe what the classifier does. Explain how someone could use it in the real world.

Your Business Proposal

The Problem

- What real-world problem are you trying to solve?

The User

- Who would use the system? Consider consumers, companies, farmers, factory workers, retailers, technicians, or public organizations.

The Solution

- How would your image recognition model be integrated into a product or service?
- Describe a simple workflow. Example: User takes a photo → Model analyzes the image → System predicts the class → Application recommends an action.

The Value

- Why would somebody use or pay for this system?
- Could it save time, reduce costs, reduce errors, automate repetitive work, improve quality, support human decisions, or provide a new service?

Part 7 — Think About the Limitations

Not all AI system works perfectly. Identify at least three possible limitations or risks of your proposed application. For each limitation, briefly explain how you could reduce the risk.

- Inaccurate predictions
- Biased or unrepresentative datasets
- Poor image quality
- Different lighting conditions
- New objects not represented in the training data
- Cost of deploying the system
- Privacy
- Human oversight
- Consequences of incorrect predictions

Final Deliverables

1. Notebook

A Jupyter / Google Colab notebook containing:

- Dataset exploration
- Image preprocessing
- Model definition
- Training
- Evaluation
- Example predictions
- Comments explaining the main steps

The notebook must be executable and understandable.

2. Short report

Prepare a report with the following structure:

1. The problem
2. The dataset
3. The classes
4. Dataset exploration
5. The model
6. Training and performance
7. Prediction examples
8. Business application
9. Limitations and risks
10. Conclusions

The objective is not to obtain the highest possible accuracy.

The objective is to demonstrate that you understand the complete process: **Problem → Data → Images → Neural Network → Training → Prediction → Evaluation → Business Application**

A relatively simple model that your group can clearly explain is better than a highly complex model used as a black box.